

# The Mathematical Solution: *w*-Descriptors & sizes

Decoupling from pure DPR and binding mathematically to the Layout Viewport

## ✓ Restoring the Connection: Predicting Layout Width

```
index.html


```

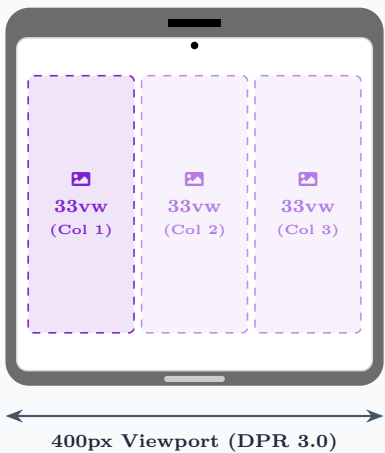
Parses sizes & srcset  
Injects Viewport & DPR

### The Preloader Math Engine:

1. Evaluate sizes → **Expected Layout Width (px)**
2. Calculate **Effective Density** for every image in srcset:

$$\text{Density} = \frac{\text{Intrinsic Width (w)}}{\text{Expected Layout Width (px)}}$$
3. Select smallest image where  $\text{Density} \geq \text{DPR}$  (or largest available)

## 📱 Scenario A Solved: Payload Slashed!



Device: Mobile (Viewport 400px, DPR 3.0)  
Grid: 3 columns (sizes="33vw")

### 🔍 Step 1: Predict Layout Width

400px viewport × 33% = **132px** expected layout width

### ⚙️ Step 2: Calculate Effective Densities

For 600w:  $600/132 = 4.5$  density ✓ ( $\geq 3.0$ )

For 1200w:  $1200/132 = 9.1$  density

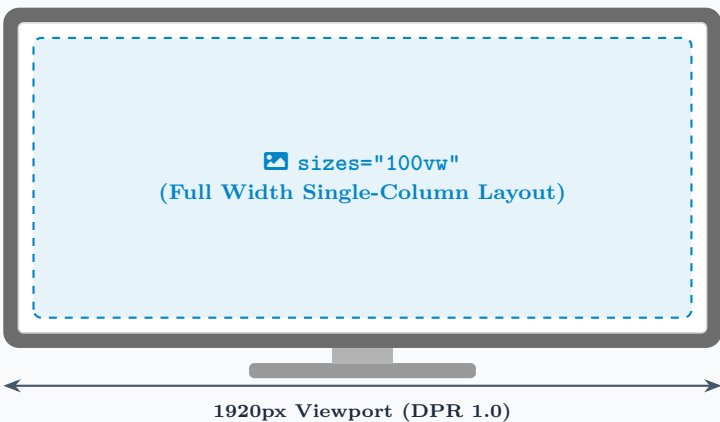
For 1800w:  $1800/132 = 13.6$  density

### ✓ Step 3: Make the Choice

Target device DPR is 3.0. The 600w image provides a layout density of 4.5, which covers the DPR completely.

**Result:** Browser dynamically downloads `img-600.jpg`.  
Pixels downloaded = 360,000px. (Down from 3.2M!)  
~90% Payload Saved from wasted bandwidth!

## 🖥️ Scenario B Solved: Crisp at any size!



Device: Desktop (Viewport 1920px, DPR 1.0)  
Grid: 1 column (sizes="100vw")

### 🔍 Step 1: Predict Layout Width

1920px viewport × 100% = **1920px** expected layout width

### ⚙️ Step 2: Calculate Effective Densities

For 600w:  $600/1920 = 0.31$  density ✗ (Fails 1.0 DPR!)

For 1200w:  $1200/1920 = 0.63$  density ✗ (Fails 1.0 DPR!)

For 1800w:  $1800/1920 = 0.94$  density ✓ (~1.0 DPR)

### ✓ Step 3: Make the Choice

Target device DPR is 1.0. The browser selects the 1800w image to provide the closest density without massive shortfall.

**Result:** Browser dynamically downloads `img-1800.jpg`.  
The image perfectly fills the massive stretched CSS grid.  
**Zero Pixelation!** (Visually; 0.94 density causes minimal 6% upscale)